Patrick Batsell

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Education

Rice University Houston, TX
BA in Operations Research and Statistics May 2027

GPA: 3.97/4.00

Relevant coursework: Generative AI for Image Data, Deep Learning, Linear Algebra, Operations Research & Optimization, Statistical Foundations of Generative AI, Stochastic Modeling, Computational Thinking, Data Science and Machine Learning.

Research Experience

Rice University Electrical and Computer Engineering Houston, TX
Undergraduate Researcher — Advisor: Dr. Richard Baraniuk Oct 2025 – Present

- Led experiments across generative models and datasets to investigate how recursive training on synthetic data changes model geometry and sample quality.
- Developed a pipeline for analyzing generator Jacobians and effective rank; connected declining effective rank to model degradation in a first-author paper presented at an ICLR 2026 workshop.
- Designed and implemented Geometrically Modified Outputs to improve generative-model self-training; reduced ImageNet256 FID from 7.32 with standard Neon to 6.25 in the best-performing configuration, leading to a first-author manuscript submitted to ICLR 2027.

Nanyang Technological University Singapore
Undergraduate Researcher — Advisor: Dr. Bihan Wen May 2025 – Sep 2025

- Proposed and developed a research project testing whether infant-scale vision-language models that recognize objects can also distinguish color, size, and texture.
- Built a controlled synthetic benchmark spanning 67 object classes and over 8,000 images; designed image-only and image–text evaluations to separate visual recognition from language grounding.
- Led experiments and wrote a first-author paper accepted at ICIP 2026 and a CVPR 2026 workshop.

Rice University Jones School of Business Houston, TX
Undergraduate Researcher — Advisor: Dr. Amit Pazgal Feb 2026 – Present

- Working on a value-decomposition framework for discrete choice modeling.

MD Anderson Cancer Center Houston, TX
Undergraduate Researcher — Advisor: Dr. Andrew Schaefer Jan 2025 – Feb 2026

- Worked to identify cost parameters for treating *H. pylori*, gastric intestinal metaplasia, and gastric cancer.

Publications

1. **Patrick Batsell**, Satoshi Tsutsui, and Bihan Wen. "Benchmarking Attribute Discrimination in Infant-Scale Vision-Language Models." *IEEE International Conference on Image Processing (ICIP)*, 2026. Also accepted to the CVPR 2026 Workshop on Cognitive Foundations for Multimodal Models.

2. **Patrick Batsell**, Thomas Walker, and Richard Baraniuk. "A Geometric Perspective on Recursive Synthetic Training." *ICLR 2026 Workshop on Deep Generative Models*, 2026.

Preprints

Patrick Batsell, Thomas Walker, and Richard Baraniuk. "Improving Generative Model Self-training with Geometrically Modified Outputs." Under review at ICLR 2027. arXiv:2609.35512.

Projects

Course Registration Optimization — Rice INFORMS Decision Lab April 2025

- Led a team developing an algorithm to optimize course registration schedules based on students' preferences.
- Evaluated schedules against the previous method of randomized allocation within each class year; achieved higher scores on our designed reward metric and higher student ratings.

Mental Disorder Classification — Rice Datathon February 2024

- Built and evaluated four models—random forest, gradient-boosted trees, SVM, and logistic regression—to identify patients with mental disorders from EEG signals, achieving 85% classification accuracy.
- Earned second place at Rice Datathon for the project.

Teaching, Leadership & Involvement

Rice University Computational and Applied Math Department Houston, TX
Teaching Assistant Jan 2026 – Present

- Supported students in class and office hours, guiding them through optimization methods and problem-solving.
- Graded assignments and reinforced core concepts through feedback and instruction.

Rice University Statistics Department Houston, TX
Lab Instructor Jan 2026 – Present

- Independently taught R programming to approximately 150 students.
- Led weekly lab sections applying R to biostatistics problems.
- Graded assignments and designed classes to reinforce core skills.

Rice University Computer Science Department Houston, TX
Teaching Assistant Aug 2025 – Dec 2025

- Held weekly office hours and guided students through problem-solving strategies in Python.
- Assisted professors with grading assignments and designing practice problems to reinforce core skills.

Rice Datathon Houston, TX
Director Oct 2024 – Feb 2025

- Secured and maintained sponsor relationships, contributing to event funding.
- Oversaw marketing, outreach, and technical support.

Awards & Honors

IEEE Signal Processing Society Student Travel Award — September 2026
CVPR Student Travel Award — June 2026
Louise J. Walsh Merit Scholarship — April 2026
President's Honor Roll — 2024–2026
Second Place, Rice Datathon — February 2024

Fellowship Applications

NSF Graduate Research Fellowship Program (GRFP)
Application submitted; decision pending.

DOE Computational Science Graduate Fellowship (CSGF)
Application submitted; decision pending.

DOD National Defense Science and Engineering Graduate Fellowship Program (NDSEG)
Application submitted; decision pending.

Skills & Interests

Technical: Python, R, Gurobi, machine learning, computer vision, PyTorch, LaTeX.
Languages: English (native), Mandarin.
Interests: Water skiing, snow skiing, fitness, weightlifting, trying new restaurants.